IE 522: Final Project

Specification

Fall 2026 · 20% of the course grade (up to 22% with extra credit)

1. Purpose

The project gives you experience working on a more realistic problem than a homework exercise, and lets you explore a topic of your own choosing. You will carry out a complete optimization study, following the workflow in the course objectives:

1. **Formulate** the problem as a mathematical program.
2. **Identify the type of model** you have formulated (LP, IP, MIP, nonlinear program, and so on).
3. **Identify a method**: an appropriate algorithm and software tool, and how they work.
4. **Implement** the model in a modeling language and solve it with a solver.
5. **Sensitivity analysis**: vary parameters that could give useful insight, and document the results.
6. **Computational study**: design experiments that test the tractability of your model and method.
7. **Write up** the study in a report.

2. Who works with whom

You may work **individually or in a group of up to three students**. This choice is open to every student, in person or remote. Working in a group is not required, but I encourage it: group work has its challenges, and it also has benefits, including getting to know other students.

A group registers on Canvas (People → Groups) before submitting, and submits one report for all its members. Every requirement, deadline, and grading criterion is the same for groups and for individuals. The expected breadth of the sensitivity analysis and the computational study is judged relative to the number of people doing the work.

3. Choosing a topic

You have two options.

Option 1 (preferred): propose your own problem. Ideally, you propose a problem, model it, and analyze it. A creative problem of your own that applies what we learned in this course earns up to **10% extra credit** (10 points on top of the 100-point report score), so your project can count for up to 22% of your course grade instead of 20%.

Your problem must meet three conditions.

- **It is realistic.** It comes from a real setting, such as your work, your research, an organization you know, or a public data source, and it uses real data or data generated in a documented, realistic way. A small example with invented numbers does not qualify.
- **You formulate it yourself.** Solving a model that is given to you in full, such as a textbook exercise, does not qualify.

- **It uses the methods of this course**: linear, integer, network, stochastic, or nonlinear programming, or a decomposition method.

If your problem overlaps with your research or with a project in another course, say so in the introduction of your report and state what work is new for IE 522. If you are unsure whether a problem is suitable, ask me before you start.

Option 2: choose a problem from the problem bank. A bank of problems will be posted on Canvas. Each problem comes with a description and a data set; you write the model yourself. Projects on bank problems are graded on the same rubric, without the extra credit. An example of a bank problem:

> **Perfection Pet Foods: empty bag inventory.** Perfection Pet Foods is a co-manufacturer of dry pet food. Each finished product is packed in its own type of bag, bought from a supplier. Decide how many bags of each type to order in each week so that all forecast production is covered at the least total cost of bags.
>
> *Data provided* (spreadsheet): for each product, the bag type, the unit weight, the supplier lead time in weeks, and the weekly forecast production in pounds; for each bag type, a price per bag that decreases with the order quantity, and a minimum order quantity.
>
> *Other requirements*: a 10% yield loss on bags during packing; a safety stock of at least 15% of each week's need, since actual needs can differ from the forecast by up to 15%; and total storage on site of at most 7 weeks of supply across all bag types.
>
> Some of the information has been simplified or modified by the company for confidentiality.

4. Deliverable

Submit on Canvas by **Wednesday, December 9, 11:59 PM**: the report as a single PDF, and the code and data. Late submissions are not accepted.

Report (at most 12 pages, excluding cover page, references and appendices)

Write for a decision maker who knows some optimization but not your problem. Use the following sections.

1. **Cover page.** Title; names, NetIDs, and section; and the UT Honor Statement, signed and dated by each author: "An essential feature of the University of Tennessee, Knoxville is a commitment to maintaining an atmosphere of intellectual integrity and academic honesty. As a student of the university, I pledge that I will neither knowingly give nor receive any inappropriate assistance in academic work, thus affirming my own personal commitment to honor and integrity."
2. **Summary.** Half a page: the problem, your approach, your main result, and your recommendation.
3. **Introduction.** The organization or setting, the problem, and the question you answer.
4. **Candidate approaches.** Two or more ways the problem could be solved, for example an exact model, a simpler model, or a heuristic, with the advantages and disadvantages of each: ease of understanding, time to implement, running time, solution quality, and cost. Justify the approach you chose.

5. **Formulation.** Sets, parameters, decision variables with units, objective, and constraints, each explained in words. State the assumptions the model makes.
6. **Model type and solution method.** Identify the type of program, the algorithm, and the software, and explain briefly how the method works on your model.
7. **Implementation.** How the model is built from the data, and how a reader can reproduce your results. Code belongs in the submitted notebook, not in the report body.
8. **Results.** The solution, in a form the decision maker can use. Verify its feasibility independently of the solver. Report the objective value and the running time. If you did not prove optimality, report a bound and the optimality gap.
9. **Sensitivity analysis.** The parameters you varied and why, the experiments, the results, and what they mean for the decision maker.
10. **Computational study.** How running time and solution quality change with instance size or with the method, and what that implies for using your approach in practice.
11. **Recommendations.** What the decision maker should do, based on your results.
12. **Limitations and future work.** What your model leaves out, and what you would do with several more months of work.
13. **What you learned.** A short paragraph on what the project taught you about optimization modeling or methods.
14. **AI use statement.** Following the course AI policy: which tools you used, for which parts of the project, and how (for example brainstorming, drafting, corrections, reviewing, code, citations). If you used none, say so.
15. **References.**

Code and data. Submit alongside the report a Jupyter notebook (or scripts) and the data files. The notebook must run from top to bottom and reproduce every number in the report.

5. Grading

The project is 20% of the course grade and is graded on the report, out of 100 points, plus up to 10 points of extra credit for a problem you propose. Each criterion is scored on four levels: *full* (all of the points), *most* (about three quarters), *partial* (about half), and *minimal* (a quarter or less).

Criterion	Full credit	Pts
Summary and problem definition	The summary states the problem, approach, result, and recommendation. The problem, scope, and assumptions are clear to a reader new to it.	10
Candidate approaches	At least two approaches compared on stated criteria; the choice is justified.	5

Criterion	Full credit	Pts
Formulation	Correct and complete. Every set, parameter, and variable is defined with units; every constraint is explained in words; the model matches the problem as described.	20
Model type and method	The type of program is identified correctly. The algorithm and software fit the model, and how the method works is explained.	10
Implementation and reproducibility	The notebook runs from top to bottom and reproduces every reported number. Data sources and preprocessing are documented.	15
Results and verification	The solution is presented in a usable form. Feasibility is checked independently of the solver. Objective value and running time are reported, with a bound and gap when optimality is not proved.	15
Sensitivity analysis	Parameters are chosen for a stated reason; experiments are designed, run, and documented; conclusions follow from the results.	10
Computational study	Running time and solution quality are measured against instance size or method, and the implications are stated.	5
Recommendations, limitations, future work	Recommendations follow from the results; limitations are stated honestly; future work is specific.	5
Writing	Organized as specified, within the page limit; tables and figures are labeled and referred to in the text.	5
Total		**100**
Extra credit (option 1 only)	Up to 10 points for your own problem: creative, realistic, with real or realistically generated data, and applying the methods of this course.	+10

Required items. A report without the signed Honor Statement or the AI use statement loses 5 points for each item missing.

6. Questions

Come to office hours or send me a calendar invite.